Shaoxuan Yin

GRAPHICS, RENDERING & AI SYSTEMS ENGINEER

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Engineer spanning real-time graphics, neural-rendering research, and AI platform work. Built a standalone Vulkan engine with hybrid deferred and hardware ray-traced paths, a 12-camera capture system and a 209GB public dataset for 3D Gaussian Splatting research, and NIUA, a personal side project: a serverless GPU platform serving seven generative models behind one API and an MCP server. Comfortable from GPU and shader code up through backend orchestration and developer-facing tooling.

TECHNICAL SKILLS

Languages: Modern C++ (C++23 / C++20), CUDA, Python, TypeScript, Rust, C#, GLSL, HLSL
Graphics / Rendering: Vulkan, OpenGL 4.3, DirectX 12 (learning), deferred + hardware ray tracing, path tracing, PBR, global illumination, OIDN / OptiX / NRD denoising
AI / Gen-media: Serverless GPU inference, image / 3D-mesh / texture / motion / music models, MCP tool-calling, agentic workflows, SDK design
Research / 3D: NeRF, 3D Gaussian Splatting, neural rendering, Nerfstudio, differentiable rendering (Taichi, NVIDIA Warp, nvdiffrast)
Systems / Tooling: CMake, Git, Docker, Linux, async job orchestration, profiling with Nsight, RenderDoc, PIX

EXPERIENCE

Freedo Technology
Feb 2026 - Present

AI Algorithm Intern | Remote

- Working on fluid simulation and differentiable rendering systems.
- Using Taichi, NVIDIA Warp, and nvdiffrast in graphics- and simulation-related workflows.

Beijing Guoyao Xintiandi Information Technology
Aug 2022 - Sep 2022

C++ Development Engineer Intern | Beijing, China

- Built a 3D visualization tool in Qt + OpenGL, improving rendering performance by 40%; also wrote a custom C++ network protocol for the data layer.

PROJECTS

NIUA (side project)
2025 - 2026

Personal side project | serverless GPU, SDK, MCP

- Agentic game-dev tool: one API over seven self-hosted open-source models (image, 3D mesh, PBR texture, music, motion) with agent-driven, multi-step asset workflows.
- Built a serverless GPU pipeline with independently deployable services and async job orchestration; cut model cold-start from ~10 min to ~20 s.
- Shipped developer-facing surfaces: an SDK, an MCP server for tool-calling agents, and Unreal/Blender plugins.

Ohao Engine
Nov 2024 - Present

C++, Vulkan, GLSL -- ~50K C++ + ~15K GLSL across 121 shaders, no editor dependency

- Standalone Vulkan rendering engine. Hybrid deferred + KHR ray-tracing paths share one scene representation, bindless materials/textures, and a TLAS/BLAS lifecycle (rebuild on geometry

change, refit on transform-only updates).
- Path tracer: multiple importance sampling (balance + power heuristic), environment-map importance sampling via precomputed marginal/conditional CDFs, Owen-scrambled Sobol QMC, variance-driven adaptive sampling, and three denoiser backends (Intel OIDN, NVIDIA OptiX, NRD).
- Deferred path: Cook-Torrance GGX, cascaded shadow maps, SSAO, screen-space reflections, subsurface scattering, TAA, bloom. Composited with ray-traced 1-bounce indirect (temporal blending) and ray-traced shadows on the same scene buffers.

Master's Thesis: Neural Rendering Dataset Collection **Mar 2025 - Dec 2025**
Linkoping University -- C++, C#, WinUI, Sapera SDK, GenICam

- Built CamMatrixCapture, a 12-camera synchronized capture system with staggered triggering, Bluetooth turntable automation, and state-machine based workflow control.
- Created and released a 209GB public dataset for NeRF and 3D Gaussian Splatting with challenging reflective, transparent, and translucent materials.
- Benchmarked NeRF vs. 3DGS and found an 11dB PSNR advantage for 3DGS on reflective surfaces.

NocturneAI (open source) **Mar 2025 - Oct 2025**
TypeScript -- ~30K LOC multi-agent CLI framework, github.com/Qervas/NocturneAI

- Multi-agent CLI framework using GitHub Copilot's API as LLM backend. ReAct loops, tool registration/execution, workflow engine.
- Clean architecture (core / application / infrastructure / presentation), AgentCoordinator for multi-agent orchestration, three modes (Agent / Ask / Edit) via ModeManager.
- Iteration loop with per-step task analysis, workflow validator for multi-step plans, plugin manager, metrics collector, background-task manager.

Path Tracer **2024**
C++, CUDA

- Built a Monte Carlo path tracer with BVH acceleration, Russian roulette termination, and multiple material models.
- Reached 40 FPS on RTX 4060 at 1280x720 through CUDA acceleration.

EDUCATION

Linkoping University **2023 - 2026**
MSc, Computer Science -- Graphics and Visualization | Linkoping, Sweden

- Graduated Feb 2026. Top grades (5/5) in Advanced Global Illumination, Modelling & Animation, Computer Games Design.

Beijing Information Science and Technology University **2018 - 2023**
BEng, Computer Science | Beijing, China

ADDITIONAL

Publication: "Skiing Action Analysis Based on OpenPose," Intelligent Computer and Applications, 2022
Languages: English (fluent), Chinese (native), French (basic)